

Md Khalequzzaman Sarker Likhon

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RESEARCH INTERESTS

Computer vision and video · Vision-language and multimodal models · LLMs and grounded generation

EDUCATION

Ahsanullah University of Science and Technology

B.Sc. in Computer Science and Engineering

Apr 2016 – Jan 2021

Dhaka, Bangladesh

- CGPA: 2.74 / 4.00
- **Thesis:** *Design of an Arrhythmia Classification Algorithm Using 2-D Convolutional Neural Networks* (Supervisor: Prof. Dr. Mohammad Shafiul Alam)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Accelx Inc.

Real-time video understanding, vision-language models and grounded LLM systems

Jan 2025 – Present

Dhaka, Bangladesh

- **Violence recognition in video.** Fine-tuned a Kinetics-pretrained R(2+1)D-18 and deployed it for real-time multi-camera monitoring. Compared it on RWF-2000 with optical-flow, pose-fusion and pose-Transformer models, and it beat all three. Also fine-tuned a YOLOv10-L weapon detector for live streams.
- **Checking what the pose models learned.** Traced their inter-person “contact” cues to keypoints missing for both people, and found the models did not track people across frames, so could not tell who did what.
- **Zero-shot fall and violence recognition with CLIP.** Built recognizers on a frozen OpenCLIP model, traced their false positives on real footage to camera viewpoint, lighting and posture, and turned these cases into hard-negative prompts, a fix that needs no retraining.
- **Grounded LLM and VLM outputs for human reviewers.** Built a pipeline where a local LLM writes alerts only from facts computed in code, and added an automatic check on every citation. Designed an OCR → CLIP → Qwen3-VL cascade that calls the VLM only for hard cases and flags weak evidence for review.

RESEARCH PROJECTS

Telemetry vs. video for automated after-action review [GitHub]

Pilot study, Sep 2026

When the same Video LLM reviews a team episode from its game log and from its video, which claims hold up?

Ran Qwen2.5-VL-7B on 15 human–human Overcooked episodes, from the game log and from video, and checked its 1,140 timestamped claims against the log. Log claims held up 81% of the time vs. 53% from video, and most video errors were real events at the wrong time. Next: blind human rating and a larger model.

DocuRoute: evaluating retrieval choices for grounded question answering [GitHub]

2026

Built an LLM router that answers questions over SEC 10-K filings with hybrid retrieval or sandboxed text-to-SQL, and an evaluation harness to compare retrieval setups. Found that a general-domain cross-encoder reranker lowered faithfulness from 0.93 to 0.77, likely due to domain shift.

Other engineering projects: Sentinel, rule-based video anomaly detection over YOLOv8 + BoT-SORT tracking at 19.7 FPS [GitHub] · SentinelRAG, self-correcting RAG with hallucination auditing [GitHub] · Research→Code, multi-agent coding with human approval [GitHub] · more at khalequzzamanlikhon.github.io/projects

MANUSCRIPTS

* equal contribution

- R. Tanvir*, **M. K. S. Likhon***, et al. *Joint Prediction of Ultra-High-Performance Concrete Properties Using a Feature-Tokenizer Transformer with Masked Multi-Task Learning*. Under review at *Applied AI Letters* (Wiley), 2026.
- **M. K. S. Likhon***, M. M. Sarker*. *Out-of-Focus Is Not Obliteration: A Perturbation Analysis of Fingerprint Alteration Detectors on Synthetic Benchmarks*. Manuscript in preparation, 2026.
Detectors reaching 0.998 macro-F1 on SOCOFing respond to editing artifacts rather than to the alteration itself.
- **M. K. S. Likhon***, M. M. Sarker*. *Does Geometry Help? A Leakage-Audited, Negative-Result Study of Cross-Attention Fusion and Taxonomy-Consistency Loss for Post-Earthquake Damage-Type Classification*. Manuscript in preparation, 2026.
Geometry fusion performs no better than a shuffled-geometry control on PEER Hub ImageNet.

TECHNICAL SKILLS

Methods	Video understanding (action recognition, pose estimation, tracking); vision-language models and Video LLMs (Qwen2.5-VL, Qwen3-VL, CLIP); object detection; transformer fine-tuning; LLM grounding and RAG; evaluation design and error analysis
Frameworks	PyTorch, Hugging Face Transformers, OpenCLIP, Ultralytics, OpenCV, scikit-learn, W&B
Engineering	Python, C/C++ , SQL, Bash; FastAPI, TorchServe, Docker, Redis, Git, Linux

SERVICE AND TEACHING

Organizing Committee, AUST Codeware (programming contest)	2019
Home tutor for secondary and higher-secondary students (grades 10–12)	2017 – 2022

LANGUAGES

Bengali (native) · English (full professional proficiency)